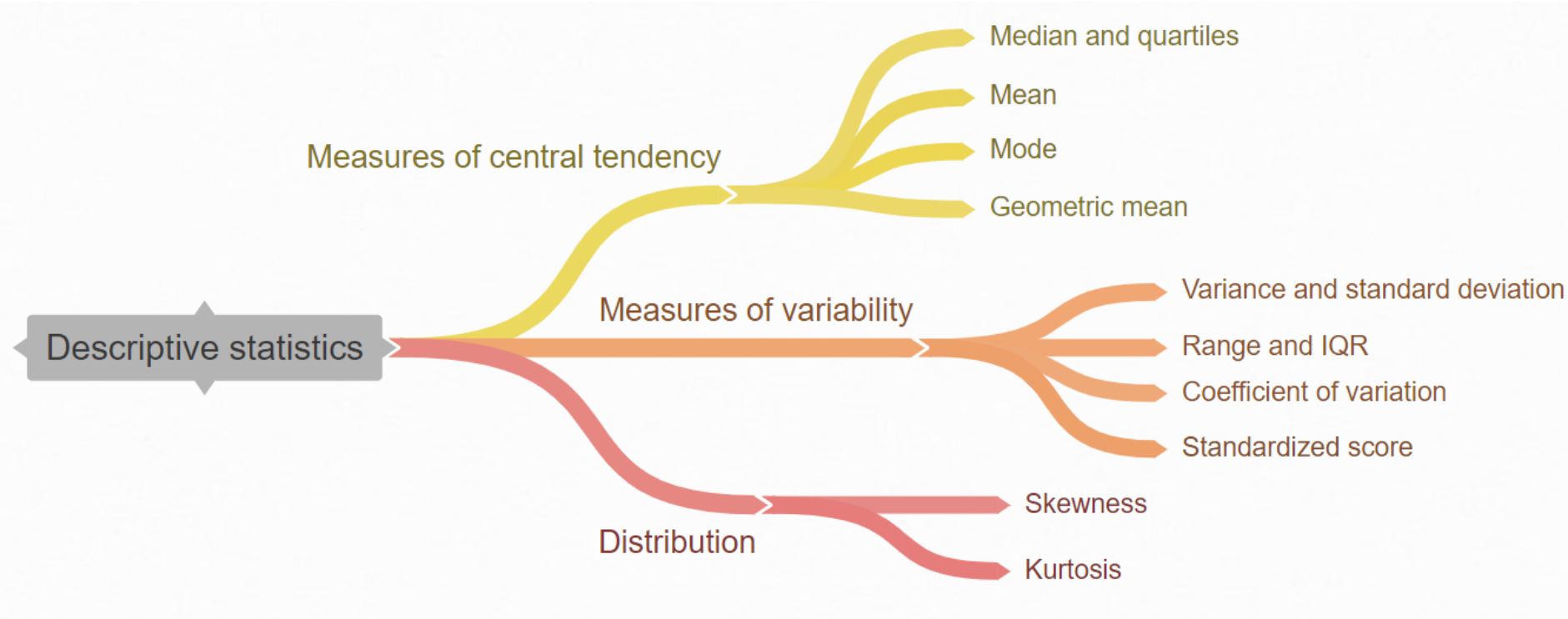
Regression

Jiayun Jin

Review


Descriptive statistics
Measures of central tendency
Median and quartiles
Mean
Mode
Geometric mean
Measures of variability
Variance and standard deviation
Range and IQR
Coefficient of variation
Standardized score
Distribution
Skewness
Kurtosis

1. Preface

2. Simple linear models
   - OLS regression
   - The goodness of fit
   - Significance test
   - Residual analysis

3. Polynomial regression

4. Multiple Linear Regression
   - The goodness of fit
   - Test of significance
   - Multicollinearity
   - Variable selection and stepwise regression

Measuring the relationship between two variables

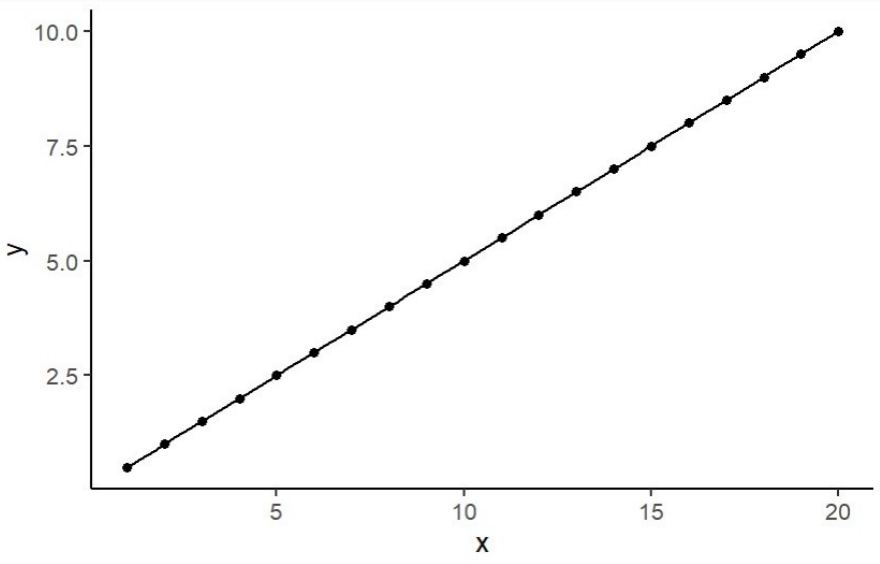
The relationship

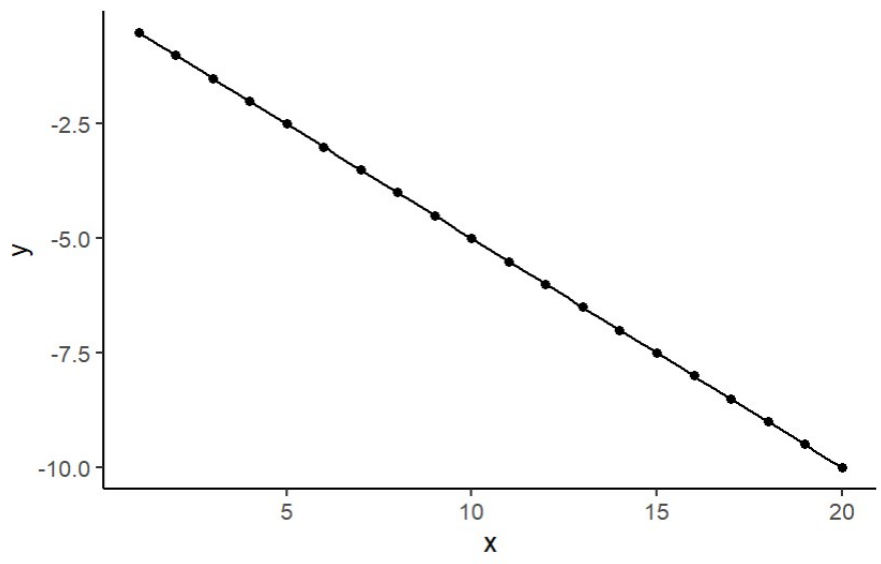
1. Not perfect but with a rule to follow:
   - income and educational level
   - child's height and parents' height
2. To be answered:
   - Is there a relationship ?
   - What kind of relationship ?
   - Strong or weak ?
   - What about the representativeness ?

Measuring the relationship between two variables

Description of the relationship: scatter plot

Deterministic (or functional) relationships

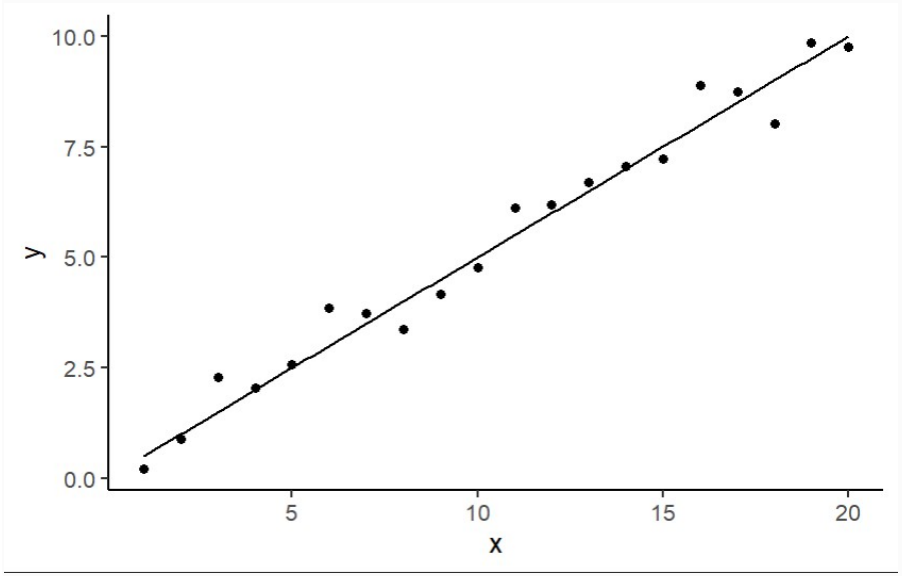


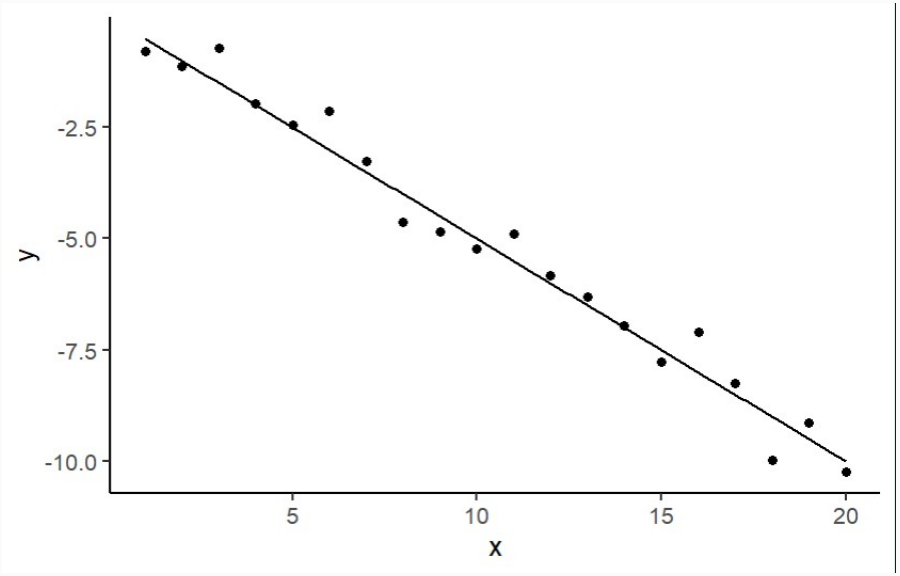


Measuring the relationship between two variables

Description of the relationship: scatter plot

statistical relationships (linear relationship):

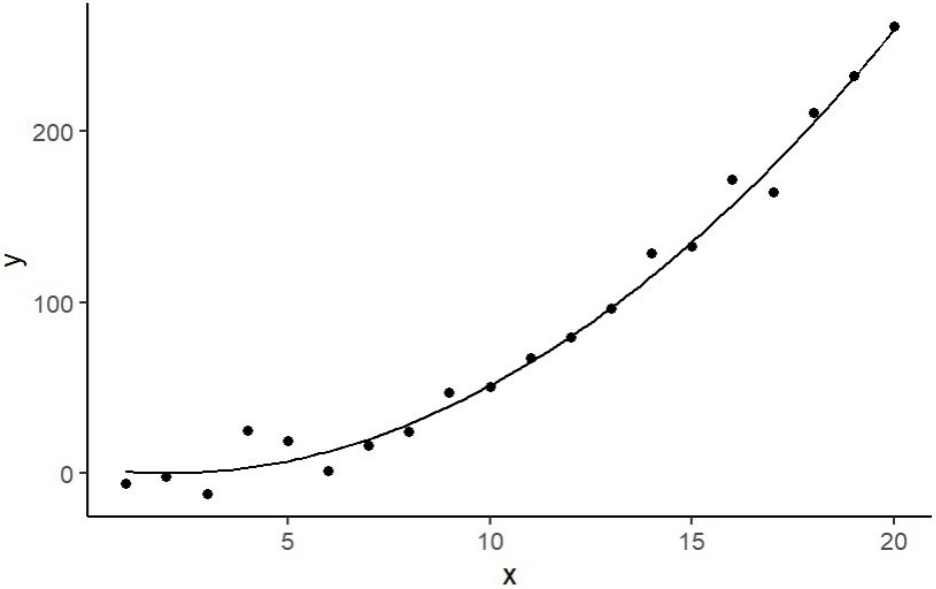




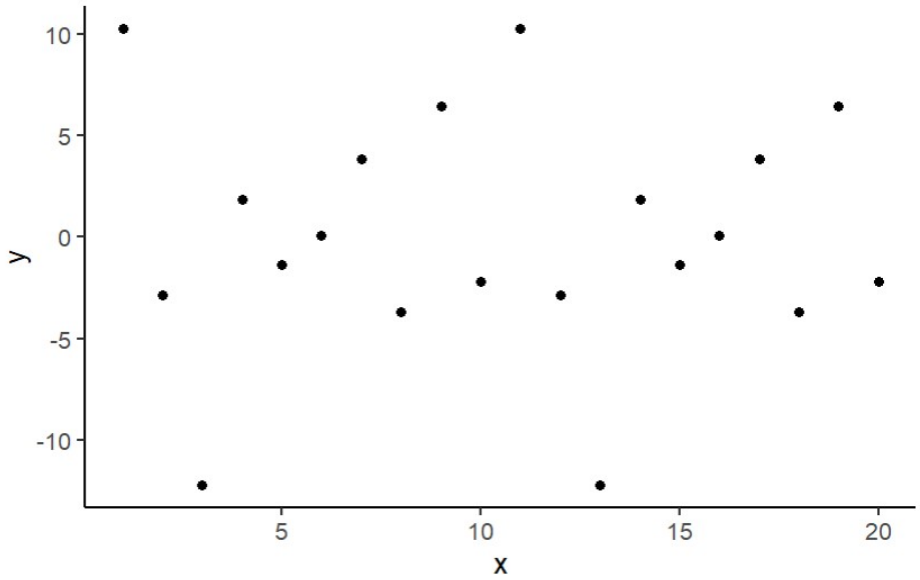
Measuring the relationship between two variables

Description of the relationship: scatter plot

non-linear:



not related:



Pearson's Correlation coefficient

A correlation coefficient reflects the strength and direction of the linear relationship between variables

$$r = \frac{cov(x,y)}{\sqrt{var(x)}\sqrt{var(y)}} = \frac{E[(x-\bar{x})(y-\bar{y})]}{\sqrt{var(x)}\sqrt{var(y)}}$$
$$= \frac{n\sum xy - \sum x \sum y}{\sqrt{n\sum x^2 - (\sum x)^2}\sqrt{n\sum y^2 - (\sum y)^2}}$$

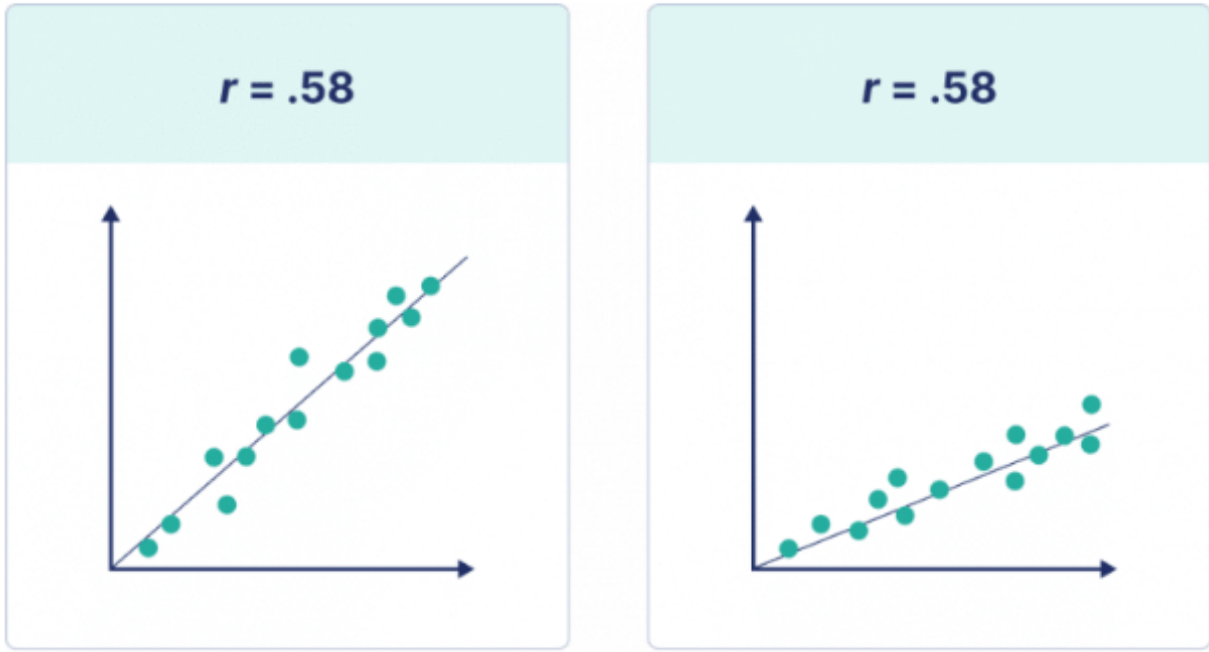
Pearson's Correlation coefficient

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$$= \frac{n\sum xy - \sum x \sum y}{\sqrt{n\sum x^2 - (\sum x)^2}\sqrt{n\sum y^2 - (\sum y)^2}}$$

- $-1 \leq r \leq 1$
- Sign
  - $r > 0$: positive
  - $r < 0$: negative
- Absolute value
  - $0 < |r| < 0.3$: weak
  - $0.3 < |r| < 0.5$: moderate
  - $0.5 < |r| < 0.7$: strong
  - $0.7 < |r| < 1$: very strong

Correlation coefficient



Note: The slope of the line does not reflect the correlation coefficient value.

Pearson's correlation coefficient

Not a good measure of correlation if:

- nonlinear relationship
- outliers
- skewed distributions
- categorical variables

Consider a rank correlation measure instead.

Non-parametric correlation coefficient

Spearman's rank correlation coefficient

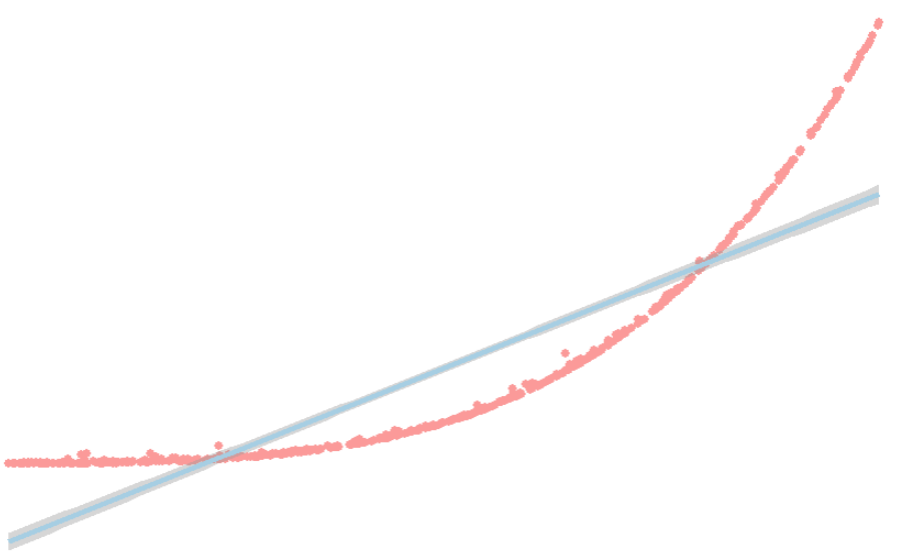
Spearman's correlation is simply the Pearson's correlation of the rankings of the raw data.

x	y	xrank	yrank
32	12	5	5
24	4	4	1
12	7	1	2
23	9	3	3
20	10	2	4

Spearman's rank correlation coefficient

$$\rho = 1 - \frac{6 \sum d_i^2}{(n^3 - n)}$$

- d_i: difference between x-rank and y-rank for each pair of data
- Monotonic relationship instead of linear relationship (fixed slope)



$$r = 0.884$$

$$\rho = 0.996$$

Correlation coefficient

Correlation does not imply causation

- Correlation: statistical association, they co-vary. This covariation isn't necessarily due to a direct or indirect causal link
- Causation: a change in one variable causes a change in another variable. A cause-and-effect.

Correlation does not imply causation

Why?



Correlation does not imply causation

Why?

1. 混杂变量 (confounding variable), e.g.,
   - 冰淇淋销量与游泳池溺水事故之间存在相关性，但实际上它们都与第三个变量——气温（夏季）有关



Correlation does not imply causation

Why?

2. 反向因果, e.g.,
   - Exercise and happiness
   - Vitamin D and depression

Correlation does not imply causation

Why?

2. 反向因果, e.g.,
   - Exercise and happiness
   - Vitamin D and depression
3. ...

Correlation does not imply causation

Why?

2. 反向因果, e.g.,
   - Exercise and happiness
   - Vitamin D and depression
3. ...

To test causation

要确定因果关系，通常需要更多的证据，包括但不限于：

- 实验设计
- 固定效应模型和工具变量分析

1 Preface

2 Simple linear models
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4 Multiple Linear Regression
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Simple linear model

Variables

- 因变量 Dependent variable：to be explained and predicted
- 自变量 Independent variable：used to explain and predict the dependent variable

Problems to solve

- Determine the mathematical function $x \rightarrow$ y
- Statistical tests
- Predictions

Simple linear model

Model

$$y = \beta_0 + \beta_1 x + \epsilon$$

- β_0 and β_1 are parameters
- $\beta_0 + \beta_1 x$: change in y due to x
- ϵ is error: the random effect on y except x

Simple linear model

Model

$$y = \beta_0 + \beta_1 x + \epsilon$$

- β_0 and β_1 are parameters
- $\beta_0 + \beta_1 x$: change in y due to x
- ϵ is error: the random effect on y except x

Assumptions

- linear relationship
- $\epsilon \sim N(0, \sigma^2)$
  - Homogeneity of variance (homoscedasticity)
  - Normality
  - Independence

Simple linear models

Regression model

$$E(y) = \beta_0 + \beta_1 x$$

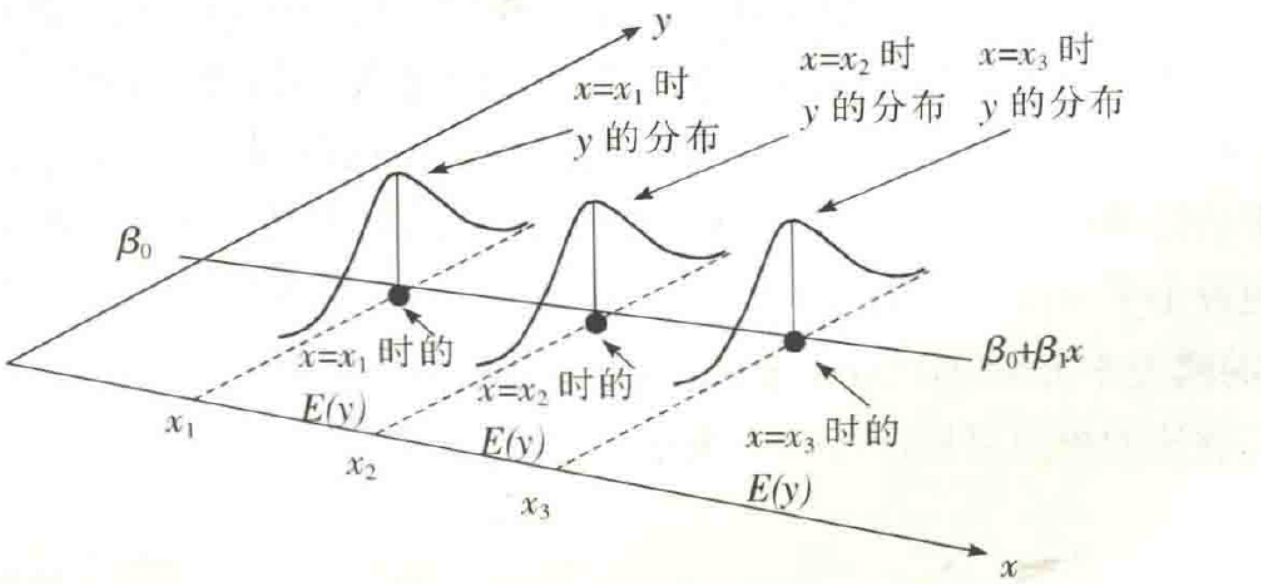
- β_0 is 截距 intercept，expected value of y when $x = 0$
- β_1 is 斜率 slope，the expected change in y due to one unit change in x

Simple linear models

Regression model

$$E(y) = \beta_0 + \beta_1 x$$

- β_0 is **截距** intercept, expected value of y when $x = 0$
- β_1 is **斜率** slope, the expected change in y due to one unit change in x



Simple linear models

Estimated regression model

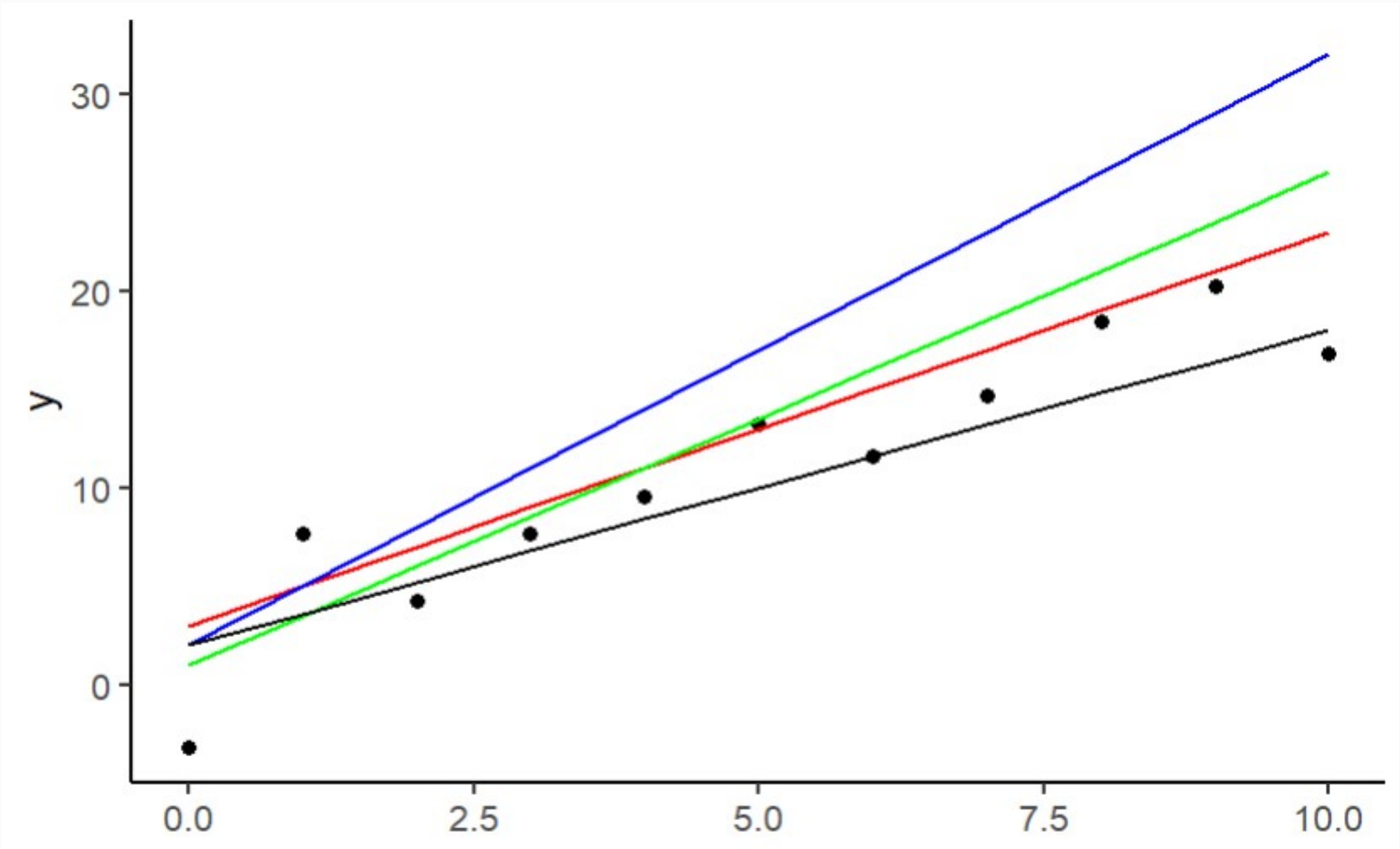
The estimated model based on sample:

$$f(x) = \hat{\beta}_0 + \hat{\beta}_1 x$$

- Based on the estimated $\hat{\beta}_0$ and $\hat{\beta}_1$, f(x) is the predicted value

Simple linear models

How to estimate β_0 and β_1



How to estimate β_0 and β_1

普通最小二乘估计 Ordinary Least Square

Find $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize:

$$\sum (y_i - f(x))^2 = \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

Therefore:

$$\hat{\beta}_1 = \frac{n \sum x_i y_i - n^2 \bar{x}\bar{y}}{n \sum x_i^2 - (n\bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

The goodness of fit

Variation decomposition

$$SST = \sum (y_i - \bar{y})^2$$
$$SSR = \sum (\hat{y}_i - \bar{y})^2$$
$$SSE = \sum (\hat{y}_i - y_i)^2$$
$$SST = SSR + SSE$$

R^2: coefficient of determination

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

Test of significance

Test of the whole model

Is the linear relationship $y = \beta_0 + \beta_1 x + \epsilon$ significant?

$$H_0 : \beta_1 = 0$$
$$H_1 : \beta_1 \neq 0$$

Test of significance

Test of the whole model

Is the linear relationship $y = \beta_0 + \beta_1 x + \epsilon$ significant?

$$H_0 : \beta_1 = 0$$
$$H_1 : \beta_1 \neq 0$$

Construction of test statistic

$$F = \frac{SSR/df_{ssr}}{SSE/df_{sse}} \sim F(k, n-k-1)$$

where k is the number of independent variables

Decisions

If $F > F_\alpha$, reject H_0, the linear relationship is significant

Significance test

Test of coefficients

Is β_1 significantly not 0 ?

$$H_0 : \beta_1 = 0$$
$$H_1 : \beta_1 \neq 0$$

Significance test

Test of coefficients

Is β_1 significantly not 0 ?

$$H_0 : \beta_1 = 0$$
$$H_1 : \beta_1 \neq 0$$

Construction of test statistic

$$t = \frac{\hat{\beta}_1}{s_{\hat{\beta}_1}} \sim t(n-k-1)$$

Decisions

If $|t| > t_{\alpha/2}$, reject H_0, x has a significant effect on y.
If $|t| > t_{\alpha/2}$, cannot reject H_0, the effect of x on y is not significant.

The evaluation of the model

Is the model appropriate?

1. Sign of the parameters
2. Significance of the parameter
3. Goodness of fit
4. Normality and homoscedasticity of ϵ

Residual analysis

The assumptions

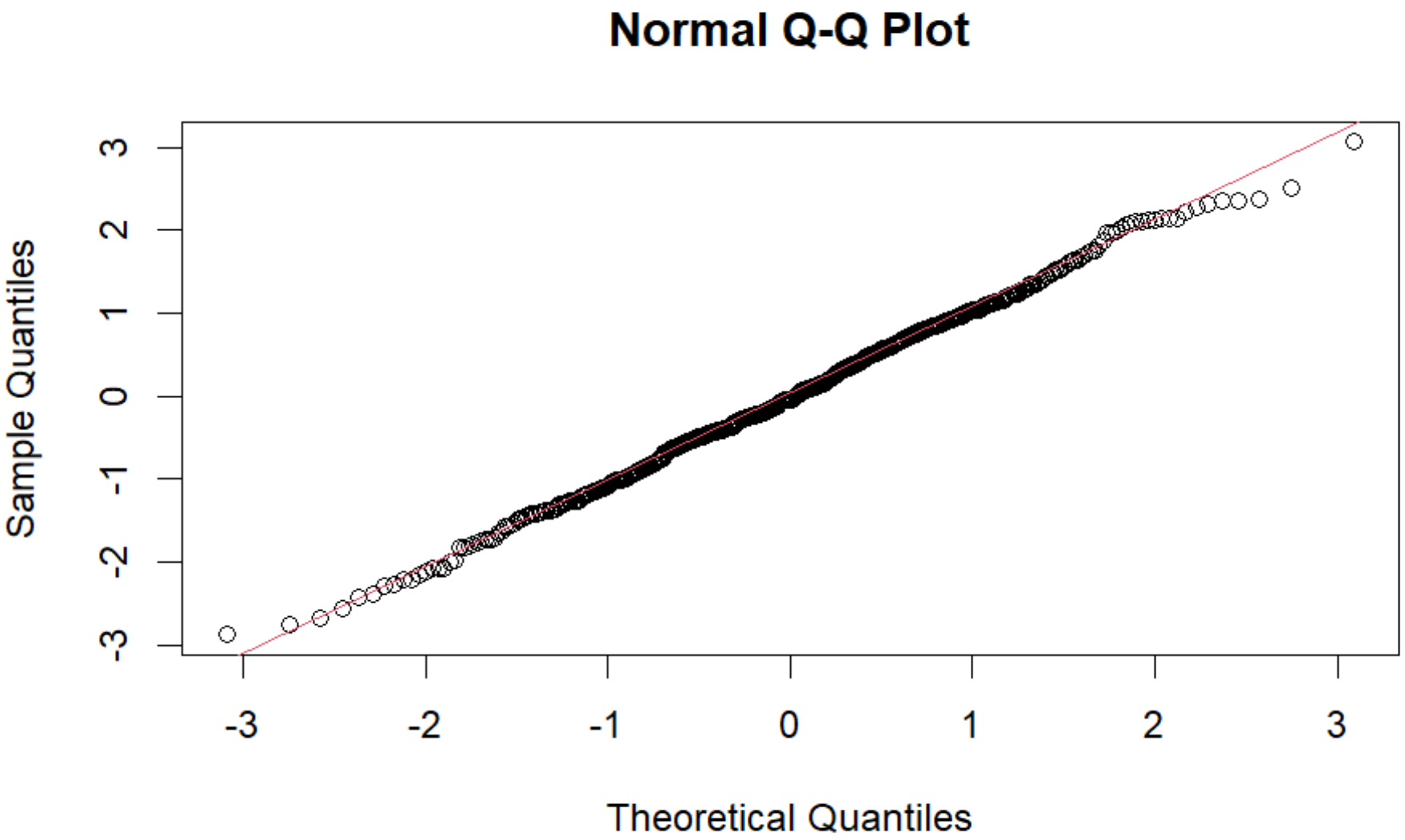
$$e_i(yi - \hat{y}_i) \text{i.i.d } \mathcal{N}(0, \sigma^2)$$

- Normality
- Equal variances
- Independence

Assumption 1: Normality

- Visually using histograms or Q-Q plots
- Formal statistical tests: e.g., Shapiro-Wilk, Kolmogorov-Smironov, Jarque-Barre, or D'Agostino-Pearson.

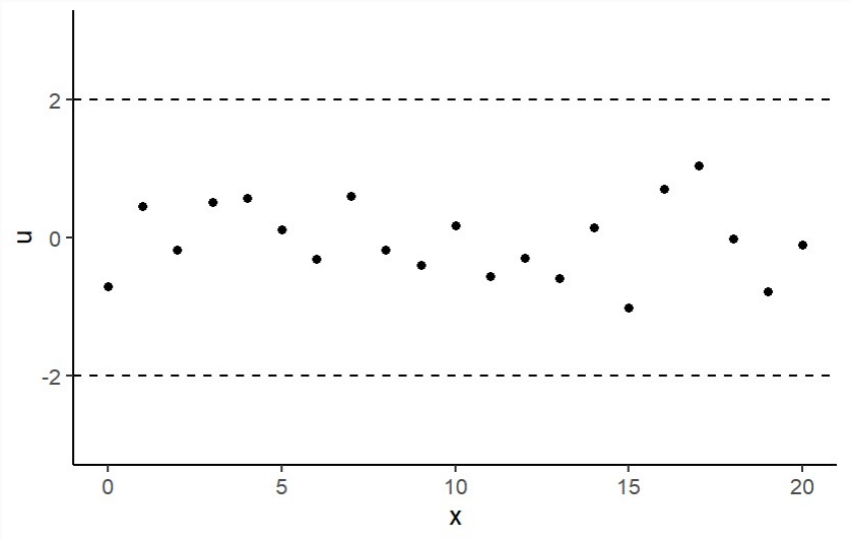
Quantile-Quantile plot

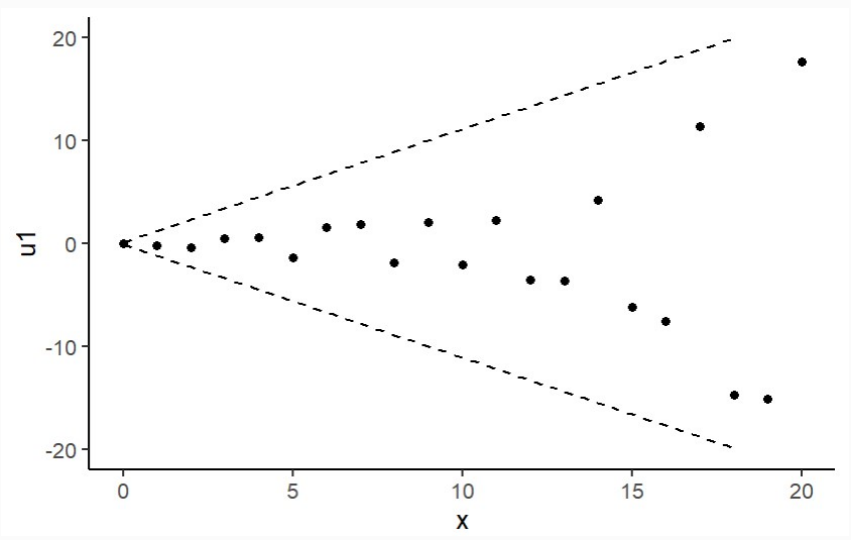


Assumption 2: Equal variance

Plot of residuals

A scatter plot displaying residuals as y-axis, and any other variables (x or y) shown as the x-axis.





1 Preface

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To account for a nonlinear relationship

Polynomial regression

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + ... + \beta_h x^h + \epsilon$$

- h refers to the degree of the polynomial
- As h gets higher, the model is more complex

Polynomial regression

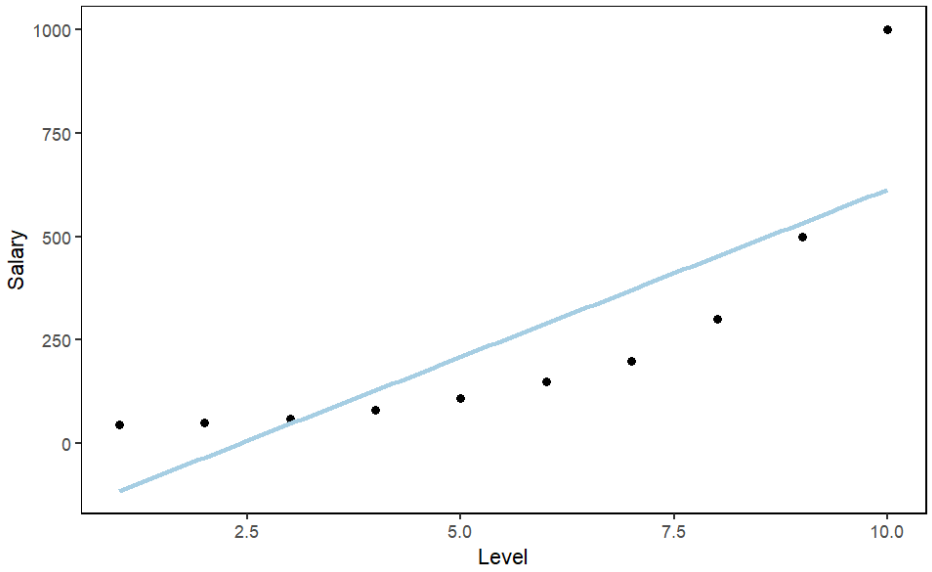
$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + ... + \beta_h x^h + \epsilon$$

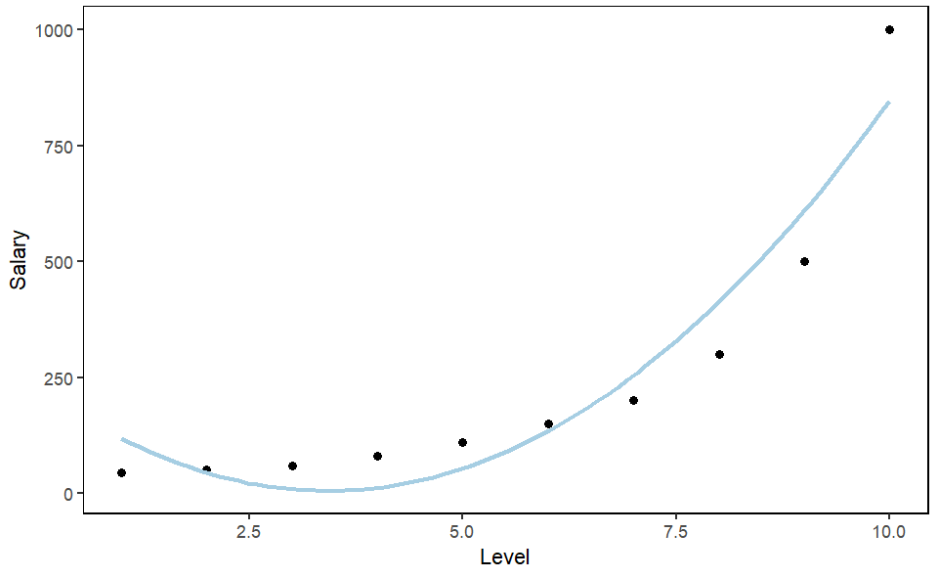
Parameter estimation: OLS

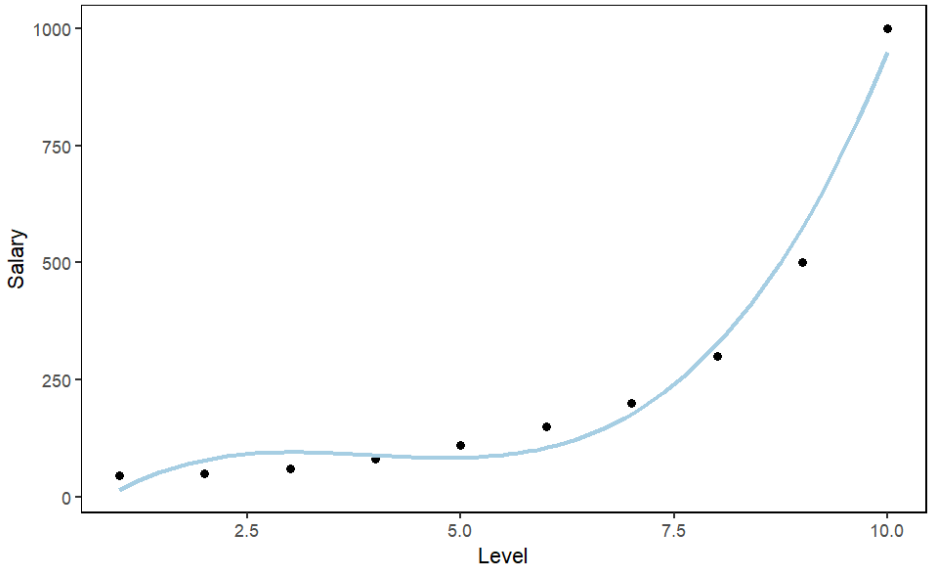
Model parameters are selected to minimize the sum of squared residuals:

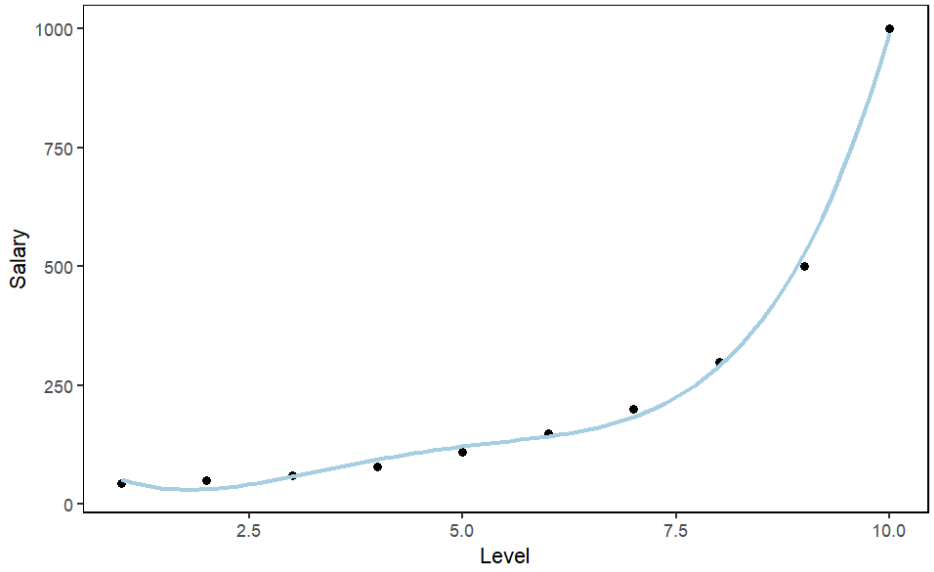
$$\sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i - \hat{\beta}_2 x_i^2 ... - \hat{\beta}_h x_i^h)^2 = \sum_{i=1}^{n} \epsilon_i^2$$

Polynomial regression

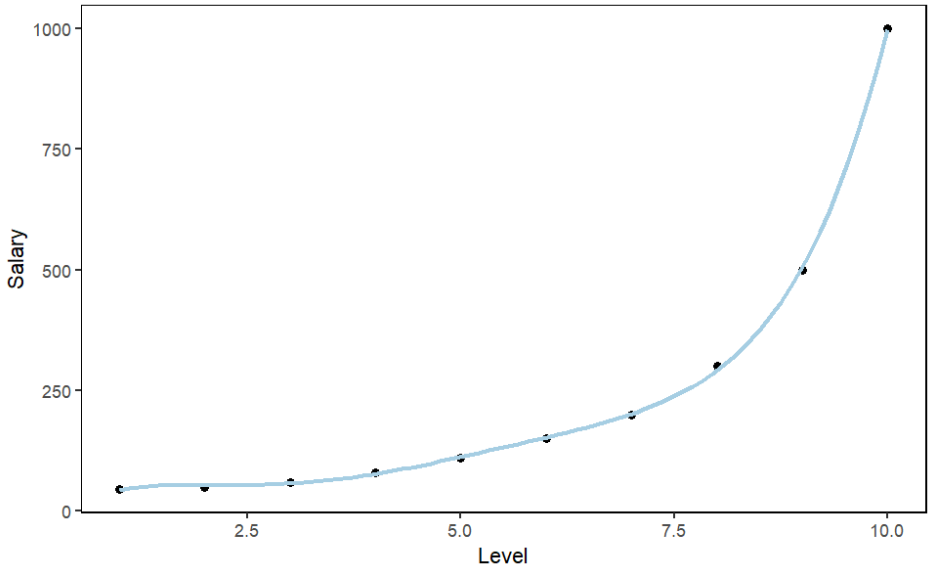


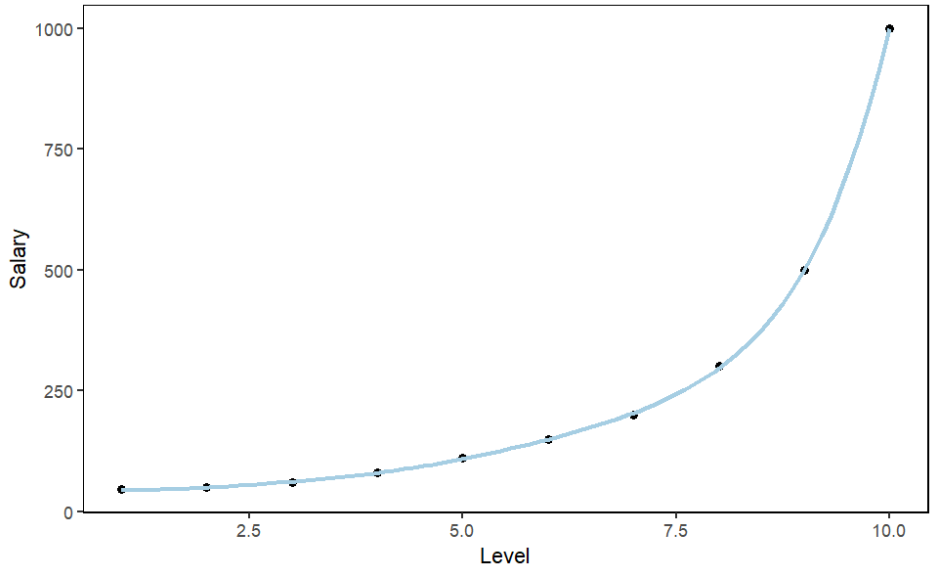




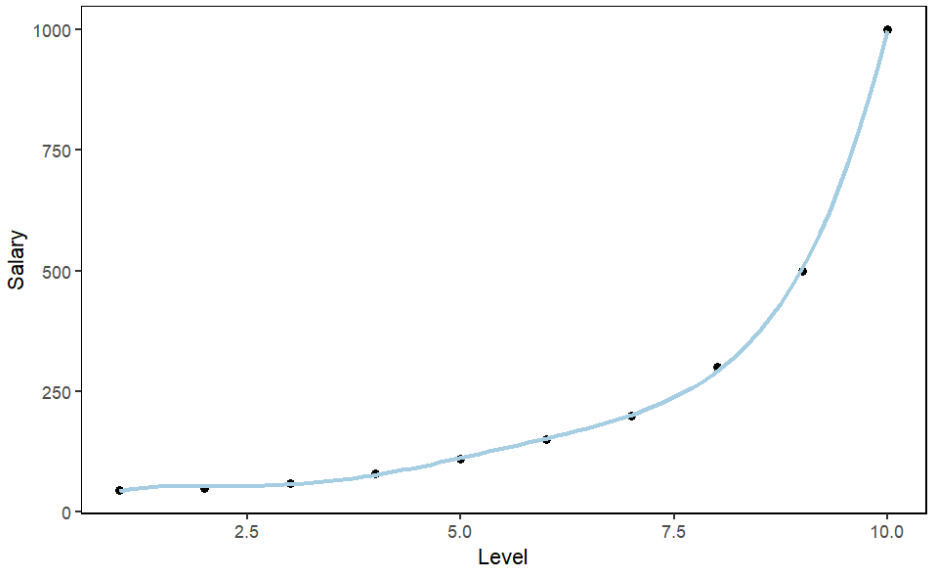


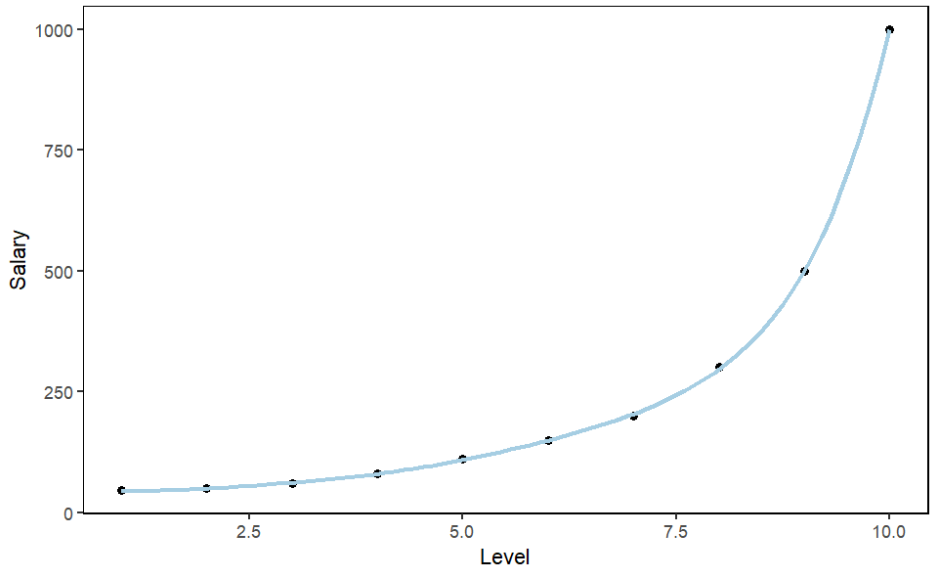
Polynomial Regression





Polynomial Regression





Overfitting

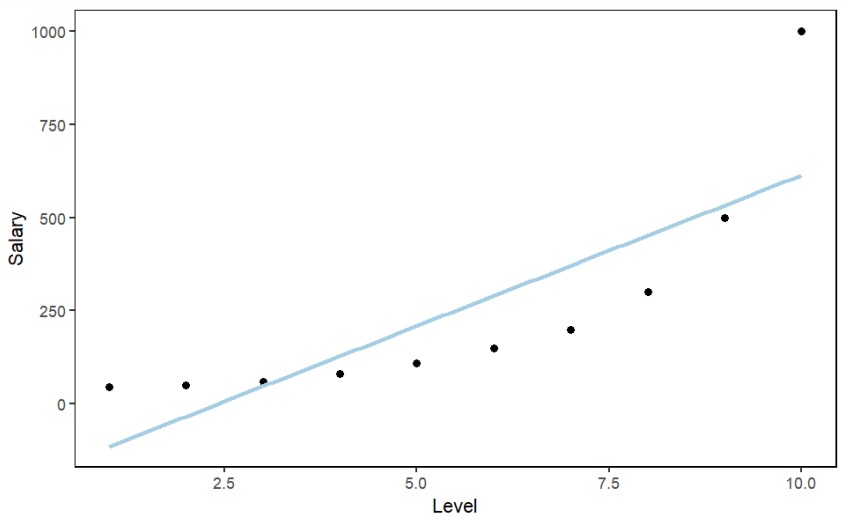
- Too closely fit on to a set of limited data points
- Perform poorly with new data points

Bias-Variance tradeoff

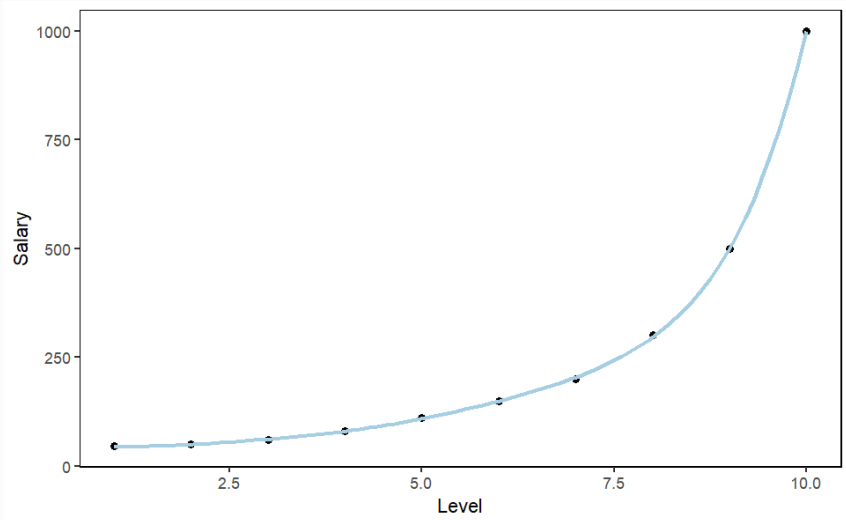
As we increase the degree of the polynomial:

- The bias decreases (the model becomes more flexible)
- The variance increases

High bias and low variance



Low bias and high variance



1. Preface

2. Simple linear models
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   - Significance test
   - Residual analysis

3. Polynomial regression

4. Multiple Linear Regression
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   - Test of significance
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Multiple Linear Regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_k x_k + \epsilon$$

where:

- β_j: The average effect of one unit increase in x_j on y, holding all other predictors fixed

OLS

Find $\hat{\beta}_0$, $\hat{\beta}_1$, ...$\hat{\beta}_k$ to minimize:

$$\sum (y_i - \hat{y}_i)^2 = \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - ... - \hat{\beta}_k x_{ik})^2$$

Hypothesis: ϵ i.i.d $N(0, \sigma^2)$

1. Normality
2. Independence
   ϵ_i is not correlated with ϵ_j
3. Equal variance

The goodness of fit

Multiple coefficient of determination

- The variance decomposition of y:

$$SST = SSR + SSE$$
$$SST = \sum (y_i - \bar{y})^2$$
$$SSE = \sum (y_i - \hat{y}_i)^2$$
$$SSR = \sum (\hat{y}_i - \bar{y})^2$$

- R^2 reflects the proportion of variance in y that can be explained by the model.

The goodness of fit

Multiple coefficient of determination

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

Adjusted multiple coefficient of determination

$$R_a^2 = 1 - (1 - R^2)(\frac{n-1}{n-k-1})$$

- Sample size n and the number of independent variables k are taken into consideration: with fixed n, R_a^2 gets smaller with increased k
- $R_a^2 < R^2$

Significance test

Significance test of the whole model

1. The statement of hypothesis

$$H_0 : \beta_1 = \beta_2 = ... = \beta_k = 0$$
$$H_1 : \text{There exists a } \beta_j \text{ that } \beta_j \neq 0$$

2. Calculate F statistic:

$$F = \frac{SSR/k}{SSE/(n-k-1)} \sim F(k, n-k-1)$$

3. Decision: If $F > F_\alpha(k, n-k-1)$, H_0 is rejected and the model is considered as significant

Significance test of the coefficients

1. The statement of hypothesis: for any β_j,

$$H_0 : \beta_j = 0$$
$$H_1 : \beta_j \neq 0$$

2. Calculate t statistic

$$t_j = \frac{\hat{\beta}_j}{s_{\hat{\beta}_j}} \sim t(n-k-1)$$

$s_{\hat{\beta}_j}$ is the standard error of the sampling distribution of $\hat{\beta}_j$:

$$s_{\hat{\beta}_j} = \frac{s_e}{\sqrt{\sum x_j^2 - \frac{1}{n}(\sum x_j)^2}}$$

3. Decision: if $|t| > t_{\alpha/2}$, H_0 is rejected and x_j is considered to have significant effect on y.

Multicollinearity

Multicollinearity

- Multicollinearity in regression analysis occurs when two or more predictor variables are highly correlated to each other

Why is it a problem

Two types of problems in general:

- The coefficient estimates of the model (and even the signs of the coefficients) can fluctuate.
- The precision of the coefficient estimates are reduced, which makes the p-values unreliable.

Multicollinearity

Detection of multicollinearity

The following situations may imply multicollinearity:

1. Check each pair of independent variable
2. Overall significant, with insignificant individual coefficient
3. The sign of the coefficient is contrary to expectation
4. Tolerance and variance inflation factor (VIF)

Detection of multicollinearity

Tolerance

The $tolerance_j$ for x_j:

$$Tolerance_j = 1 - R_j^2$$

- R_j^2 is the coefficient of determination when x_j is predicted by other independent variables. A small tolerance refers to a severe multicollinearity 。
- Typically, when it is smaller than 0.1, we consider a severe multicollinearity.

Detection of multicollinearity

Variance inflation factor(VIF)

$$VIF_j = \frac{1}{1 - R_j^2}$$

- $VIF_j = 1$: No correlation between x_j with any other predictor variables
- $1 < VIF_j < 5$: Moderate correlation between x_j with other predictors, but not severe enough to require attention
- $5 < VIF_j < 10$: Attention required
- $VIF_j > 10$: Severe multicollinearity.

Multicollinearity

How to resolve multicollinearity

- Do nothing
  - If there is only moderate multicollinearity
  - If you are interested in a predictor variable in the model that doesn't suffer from multicollinearity
  - If the goal is to make predictions and you're not concerned with understanding the exact relationship between the predictor variables and response variable
- Obtain more data
- Remove one or more of the highly correlated variables
- Ridge regression, partial least squares (PLS) regression or principal component analysis
- Variable selection

Variable selection and stepwise regression

向前逐步选择 Forward stepwise selection

- Fit every possible one-predictor model.
- Find the best significant model and the corresponding x_i, and include it first into the model
- With x_i included, fit every possible two-predictor model (k-1). Find the best significant model and include the corresponding x_j.
- Until no improvement of the model can be achieved with extra variables included.

Drawback

Once included, forever safe

Variable selection and stepwise regression

向后逐步选择 Backward stepwise selection

- Fit a model using all k predictors.
- Fit k-1 models with one variable removed at a time, and pick the best model
- Repeatedly remove variables one at a time until every variable is significant

Drawback

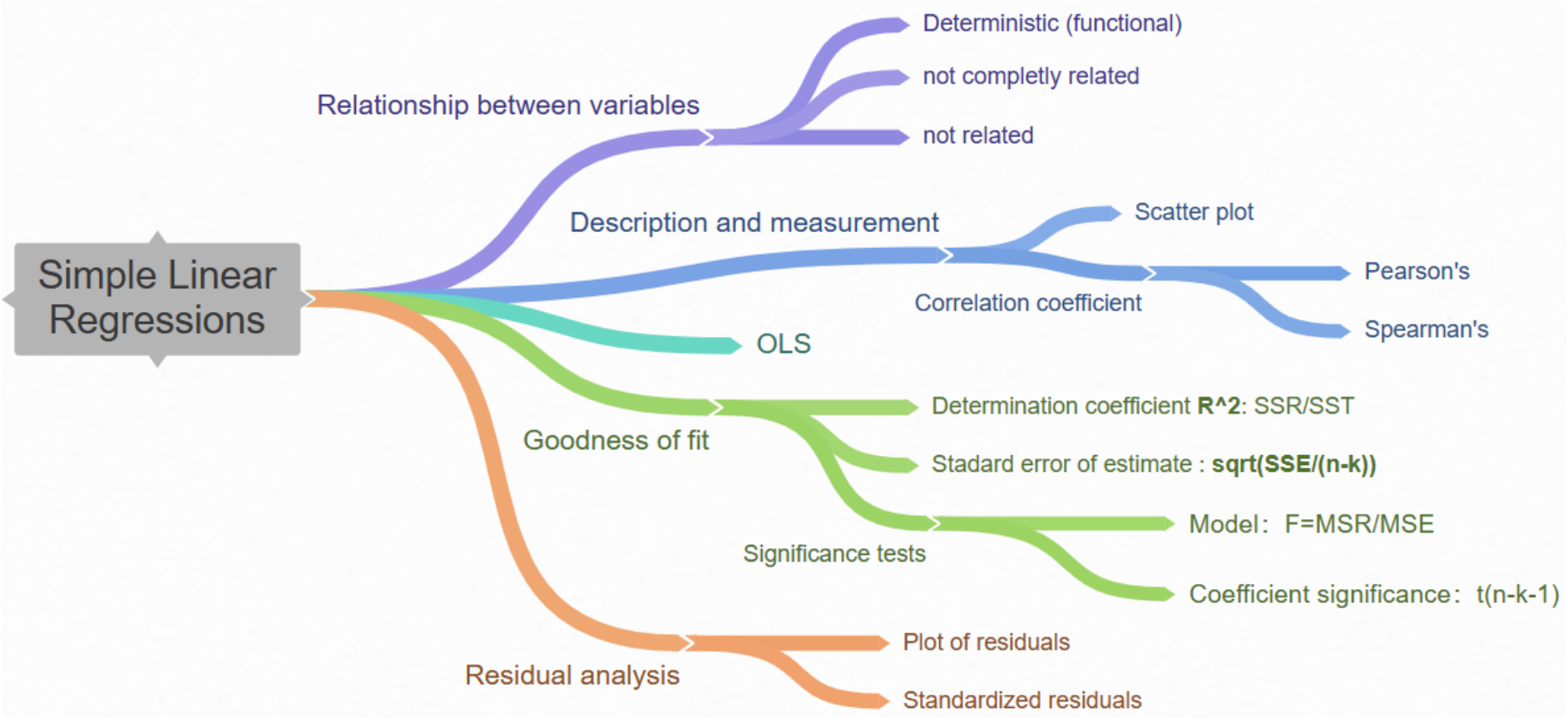
Once removed, forever goodbye

Variable selection and stepwise regression

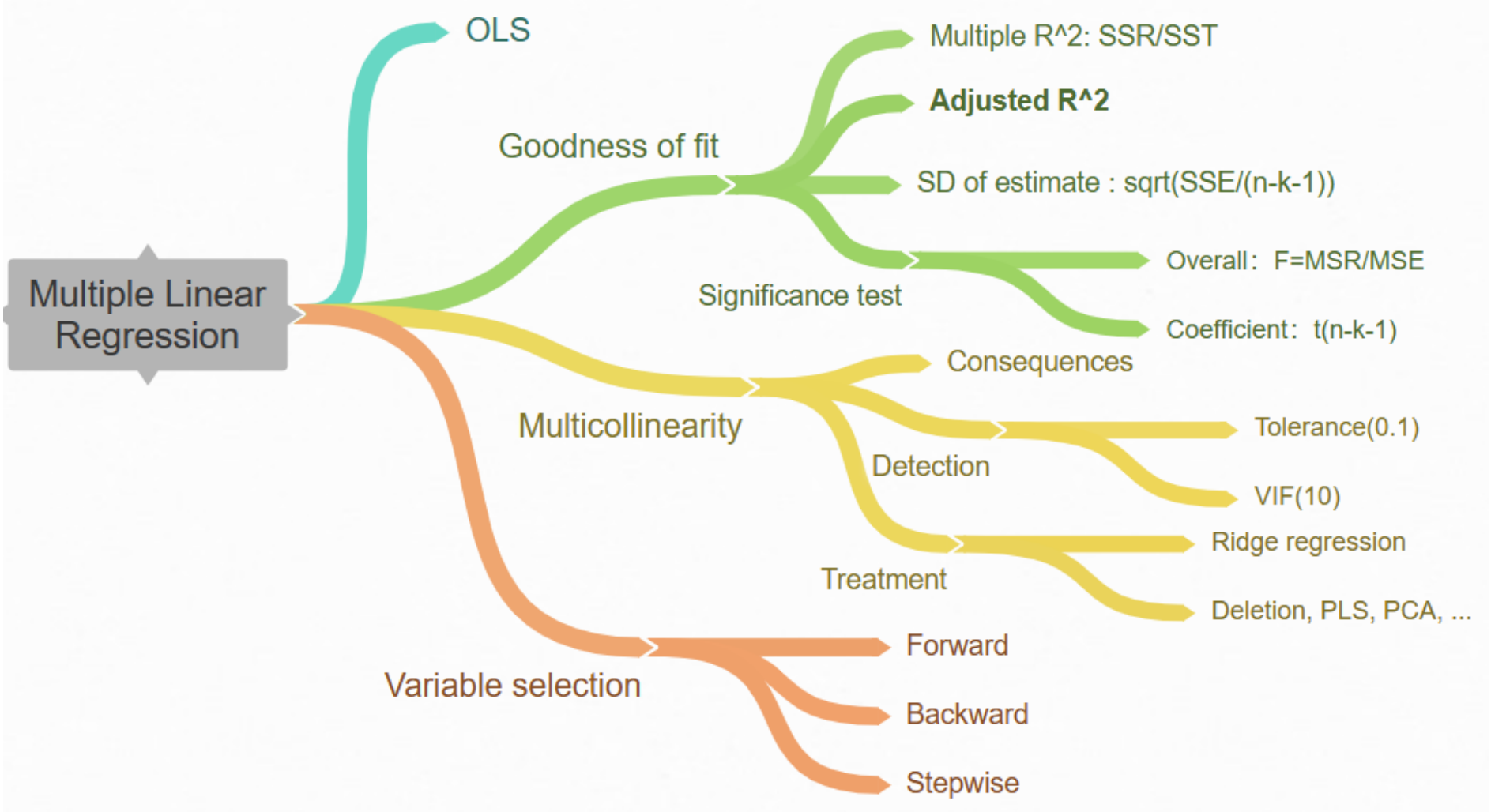
(Both-direction) Stepwise regression

- Combination of forward and backward
- Add predictors to the model sequentially
- Removed any predictors that no longer provided an improvement in model fit.

Summary


Simple Linear Regressions
Relationship between variables
Deterministic (functional)
not completly related
not related
Description and measurement
Scatter plot
Correlation coefficient
Pearson's
Spearman's
OLS
Goodness of fit
Determination coefficient R^2: SSR/SST
Stadard error of estimate : sqrt(SSE/(n-k))
Significance tests
Model: F=MSR/MSE
Coefficient significance: t(n-k-1)
Residual analysis
Plot of residuals
Standardized residuals

Summary


Multiple Linear Regression
OLS
Goodness of fit
Multiple R^2: SSR/SST
Adjusted R^2
SD of estimate : sqrt(SSE/(n-k-1))
Significance test
Overall: F=MSR/MSE
Coefficient: t(n-k-1)
Multicollinearity
Consequences
Detection
Tolerance(0.1)
VIF(10)
Treatment
Ridge regression
Deletion, PLS, PCA, ...
Variable selection
Forward
Backward
Stepwise